

Lab 05 - Modbus/TCP Routing Between Subnets



Prof. Dr. Luiz F. Freitas-Gutierrez

luiz.gutierrez@ufsm.br

linkedin.com/in/lffreitas-gutierrez



Lattes

ORCID

SciProfiles

Scopus

ResearcherID

substationworm

LFFreitasGutierrez

Tasks

- **1** Verify the IP addresses assigned to the otlab-student workstation.
- **2** Determine the subnet ranges of the networks in which the otlab-student workstation is deployed.
- **3** Discover the IP address, MAC address, and vendor information of other active hosts within the subnets. Hint: Two industrial devices belong to the same vendor.
- **4** Identify open ports and available services on both OT-ICS hosts over TCP and UDP protocols.
- **5** Intercept the communication between the OT-ICS hosts and determine the secret message being transmitted. Hint: The message must be decoded.

Tools

- These are the tools available on the otlab-student workstation for completing OTLab 05: ifconfig, masscan, netdiscover, nmap, and tcpdump.

Nomenclature

- ICS: Industrial control system.
- IP: Internet protocol.
- MAC: Media access control.
- OT: Operational technology.
- TCP: Transmission control protocol.
- UDP: User datagram protocol.